

JEFFREY (JEB) SONG

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Department of Physics, University of Washington

EDUCATION

University of Washington (UW)

2023-2025

B.Sc. in Physics (Honours), B.Sc. in Mathematics, 3.94/4.00.

National University of Singapore (NUS)

2021-2023

B.Sc. in Physics, 4.36/5.00. NUS College Programme (Honours College).

PUBLICATIONS

J. Z. Song, H. Ha, W. W. Ho, and V. B. Bulchandani, *A theory of quasiballistic spin transport*, arXiv:2503.15756, 2025. <https://arxiv.org/abs/2503.15756>

RESEARCH EXPERIENCE

UW Assoc. Prof. Mark Rudner Group ————— *Variational Cooling* *April 2025 - Present*

Investigating variational cooling for near-term quantum computers with IBM quantum backends. Currently preparing manuscript (estimated October).

Rice University Asst. Prof. Vir Bulchandani Group ————— *Quasi-Ballistic Spin Transport* *Sep 2024 - Present*

Analyzing regimes for long-lived ballistic modes of spin on anisotropic XXZ spin-1/2 chains. Contributed to analytical results and developed numerical simulations of these results. Our manuscript is currently under review in PRB.

UW Assoc. Prof. Samu Taulu Group ————— *EEG Calibration via Quasi-Static Models* *June 2024 - Present*

Developed and optimised EEG calibration algorithms informed by electromagnetic models, comparing both local and global approaches in FieldTrip. Simulated and analysed source localisation techniques. Findings are being prepared for publication.

NUS Asst. Prof. Ho Wenwei Group ————— *1D Measurement-Based Quantum Computation* *Jan 2023 - Present*

Studied 1D Measurement-Based Quantum Computation (MBQC) using tensor networks for my UROPS project. Reproduced the results of a novel 1D MBQC scheme using Python and extended their findings to accommodate higher logical qubit counts and qutrits. My UROPS report was nominated for the best UROPS award by the Physics Department.

UW Prof. Arka Majumdar Group ————— *Optical Machine Learning* *Aug 2023 - Feb 2025*

Explored multi-input reservoir computing (a form of recurrent neural networks) using optical spiral waveguides. Developed simulation scripts in Lumerical (varFDTD and FDE) to design geometries and process results. Worked on simulating input data for the etched device.

NUS Assoc. Prof. Kuldip Singh Group ————— *Foundations of Quantum Mechanics* *Jan 2023 - May 2023*

Explored ψ -ontological models, Bell's inequality, and the Pusey-Barrett-Rudolph theorem for directed reading project. Completed a project paper and presentation on limitations of such models.

NUS Asst. Prof. Lee Chinghua Group ————— *Non-Hermitian Quantum Physics with Qiskit*
May 2022 - Feb 2023

Simulated quantum systems using IBM's Qiskit package. Reproduced the SSH model, implemented trotterization for arbitrary unitaries, and decomposed non-Hermitian Hamiltonians.

NUS CQT Asst. Prof. Loh Huanqian Lab ————— *Lasers for Neutral Atom Quantum Computing*
Feb 2022 - June 2022

Built an intensity servo system and constructed optical setups involving AOMs and DDS for precise control. Presented an internal talk on error correction techniques for laser servos.

ACTIVITIES

UW Undergraduate Physics Mentoring Program *Jan 2024 - Present*
Mentor

Provided academic and personal support to three mentees, facilitating bi-weekly meetings and participating in academic and social events.

UW Society for Physics Students (SPS) *September 2023 - Present*
Member

Participated in weekly talks by guest professors, attended various seminars and events hosted by UW SPS. Hosted an information session for pre-majors on physics.

NUSC Philosophy Club *June 2022 - June 2023*
Founder

I organised bi-weekly philosophy discussions, with a focus on the philosophical aspects of science and math. I successfully organized several talks and lectures by philosophy professors, providing an opportunity for students to learn from experts in the field and expand their understanding of philosophy.

NUS Physics Society *August 2021 - August 2022*
Events Director

I helped organise a successful physics mentor-mentee program. Furthermore, I was instrumental in jointly planning and organizing online meetups for physics students, providing opportunities for networking and learning beyond the classroom.

AWARDS

Mary Gates Research Scholarship *2024*

Guy Pitzel Scholarship (Departmental Honours) *2024*

Goldwater Scholar UW Nominee *2024*

CRISP Award for Best UROPS Nominee *2023*

PRESENTATIONS

Improving EEG calibration using electrostatic multipole moments
UW Mary Gates Undergraduate Research Symposium *2025*

A Theory of Quasiballistic Spin Transport
UW SPS Research Symposium for Physical Sciences *2025*

TECHNICAL SKILLS

Technical Skills Python (Proficient), Linux (Ubuntu/Fedora), Java, PyTorch, Lean, Lumerical, Qiskit, Matlab, Bash/Zsh Scripting, C